

Divisors and unique factorizations of Carlitz binomial polynomials

September 8, 2026

Abstract

Let $A = \mathbb{F}_q[t]$. We classify the divisors in $\text{Int}(A)$ of every finite product of the basic Carlitz binomial polynomials β_{q^s} . Each divisor is, up to a unit, a subproduct of the given basic factors. Thus the multiplicative monoid generated by these factors is divisor-closed and factorial. In particular, every power of every Carlitz binomial polynomial has a unique factorization into irreducibles, answering a question of Tichy and Windisch. The proof combines a simultaneous-minimum identity for weighted products over the sets of polynomials of bounded degree with an extraction argument at a prime of prescribed degree.

1 Introduction and main theorem

Fix a prime power q , and write $A = \mathbb{F}_q[t]$ and $K = \mathbb{F}_q(t)$. The ring of integer-valued polynomials on A is

$$\text{Int}(A) = \{H \in K[X] : H(A) \subseteq A\}.$$

Here t and X are independent indeterminates. Carlitz constructed a regular A -module basis of this ring that is analogous to the ordinary binomial basis of $\text{Int}(\mathbb{Z})$; we follow the notation and normalization in Tichy–Windisch [2].

Set $V_0 = \{0\}$, $\psi_0(X) = X$, and $F_0 = 1$. For $s \geq 1$, let

$$\begin{aligned} V_s &= \{f \in A : \deg f < s\}, & \psi_s(X) &= \prod_{f \in V_s} (X - f), \\ F_s &= \prod_{i=1}^s (t^{q^i} - t)^{q^{s-i}}, & B_s(X) &= \frac{\psi_s(X)}{F_s}. \end{aligned} \tag{1}$$

We also put $B_0 = X$. The convention $\deg 0 = -\infty$ is used. If $k = \sum_{i \geq 0} \alpha_i q^i$ is the base- q expansion of a nonnegative integer, then its Carlitz binomial polynomial is

$$\beta_k(X) = \prod_{i \geq 0} B_i(X)^{\alpha_i}. \tag{2}$$

In particular, $\beta_{q^s} = B_s$.

Tichy and Windisch proved that the B_s are irreducible in $\text{Int}(A)$ and that each is absolutely irreducible: every power of an individual B_s factors uniquely into irreducibles [2, Theorem 1.2]. Their Question 3.1 asks whether this uniqueness persists for powers of the general polynomial (2). We prove the following stronger divisor statement.

Theorem 1.1. *Let $a_0, \dots, a_s \in \mathbb{N}_0$, and put*

$$P = \prod_{i=0}^s B_i^{a_i}.$$

An element $H \in \text{Int}(A)$ divides P in $\text{Int}(A)$ if and only if

$$H = u \prod_{i=0}^s B_i^{b_i} \quad \text{for some } u \in \mathbb{F}_q^\times \text{ and integers } 0 \leq b_i \leq a_i. \quad (3)$$

The unit u and the exponent vector (b_i) are unique.

Corollary 1.2. *For every $k \geq 1$ and $m \geq 1$, the polynomial β_k^m has a unique factorization into irreducibles in $\text{Int}(A)$, up to order and associates. If $k = \sum_i \alpha_i q^i$, this factorization is*

$$\beta_k^m = \prod_i B_i^{m\alpha_i}.$$

The mixed-product assertion is stronger than absolute irreducibility of the separate B_i . The proof must show that extracting a highest-level basic factor from an arbitrary divisor preserves integer-valuedness at every prime. Section 2 provides the required fixed-divisor identity; Section 3 performs the extraction and proves the theorem. The argument also recovers the known irreducibility of the B_i .

2 A simultaneous-minimum identity

For a monic irreducible polynomial $\pi \in A$, let v_π be the corresponding valuation on K , normalized by $v_\pi(\pi) = 1$, with $v_\pi(0) = +\infty$. If $R \in A[X]$ is nonzero, write

$$\mu_\pi(R) = \min_{x \in A} v_\pi(R(x)).$$

This minimum is finite: A is infinite, and R has only finitely many roots. It is the π -adic valuation of the fixed divisor of R .

The residue counts for V_s and the formula for $v_\pi(F_s)$ used below are classical; see Wagner [3, p. 51]. We require a point that simultaneously minimizes the valuation of ψ_s and of an arbitrary product of its linear factors with nonnegative multiplicities.

Lemma 2.1. *Let $s \geq 1$, let $\pi \in A$ be monic irreducible of degree d , and put*

$$E_s(\pi) = \sum_{j=1}^{\lfloor s/d \rfloor} q^{s-jd}.$$

Let

$$R(X) = \prod_{f \in V_s} (X - f)^{r_f}, \quad r_f \in \mathbb{N}_0.$$

Then, for every $c \in \mathbb{N}_0$,

$$\mu_\pi(\psi_s^c R) = cE_s(\pi) + \mu_\pi(R). \quad (4)$$

Moreover,

$$\mu_\pi(\psi_s) = E_s(\pi) = v_\pi(F_s). \quad (5)$$

Proof. Write $s = kd + r$, where $0 \leq r < d$. For $1 \leq j \leq k$, the reduction map

$$V_s \longrightarrow A/(\pi^j)$$

is surjective, and each fiber has q^{s-jd} elements. Indeed, division by the monic polynomial π^j expresses a polynomial of degree less than s uniquely as a remainder of degree less than jd plus π^j times a polynomial of degree less than $s - jd$. Consequently, for every $x \in A$,

$$\begin{aligned} \sum_{f \in V_s} \min\{v_\pi(x - f), k\} &= \sum_{j=1}^k \#\{f \in V_s : x \equiv f \pmod{\pi^j}\} \\ &= E_s(\pi). \end{aligned} \tag{6}$$

Empty sums have value zero, so this also holds for $k = 0$.

The residue class $x + \pi^k A$ contains exactly q^r elements of V_s . For $k = 0$ this says simply that $|V_s| = q^s = q^r$. These elements occupy at most q^r of the q^d residue classes modulo π^{k+1} lying above x modulo π^k . Since $q^r < q^d$, choose $y \in A$ in one of the unoccupied classes. Then $y \equiv x \pmod{\pi^k}$, and for every $f \in V_s$,

$$v_\pi(y - f) = \min\{v_\pi(x - f), k\}. \tag{7}$$

For valuations less than k , this follows from the congruence between x and y ; for valuations at least k , it follows from the choice of the class of y modulo π^{k+1} .

Equation (6) now gives

$$v_\pi(\psi_s(y)) = E_s(\pi), \quad v_\pi(R(y)) \leq v_\pi(R(x)).$$

It also shows that $v_\pi(\psi_s(x)) \geq E_s(\pi)$ for all $x \in A$. Start with an x attaining $\mu_\pi(R)$. The resulting y attains this same minimum and also minimizes ψ_s . Hence

$$\mu_\pi(\psi_s^c R) = cE_s(\pi) + \mu_\pi(R).$$

Taking $R = 1$ and $c = 1$ proves the first equality in (5).

The polynomial π divides $t^{q^i} - t$ exactly when $d \mid i$, and then its multiplicity is one. This follows from the description of the roots of $t^{q^i} - t$ as $\mathbb{F}_{q^i}$ and from its derivative -1 . Applying v_π to the product defining F_s gives the second equality in (5). $\square$

Corollary 2.2. *Every B_s belongs to $\text{Int}(A)$. More generally, if $C \in K^\times$, $s \geq 1$, $c \in \mathbb{N}_0$, and R is as in Lemma 2.1, then*

$$C\psi_s^c R \in \text{Int}(A) \iff CF_s^c R \in \text{Int}(A). \tag{8}$$

Proof. An element $z \in K$ belongs to A if and only if $v_\pi(z) \geq 0$ for every monic irreducible $\pi \in A$. Therefore, for a nonzero polynomial $Q \in A[X]$,

$$CQ \in \text{Int}(A) \iff v_\pi(C) + \mu_\pi(Q) \geq 0 \text{ for every } \pi.$$

Lemma 2.1 makes these inequalities identical for the two polynomials in (8). It also proves $\psi_s/F_s \in \text{Int}(A)$. The case $B_0 = X$ is immediate. All primes in the criterion are finite primes of A ; no condition at the infinite place is imposed. $\square$

3 Extraction and divisor classification

Lemma 3.1. *Let $s \geq 1$, let $a_0, \dots, a_s \in \mathbb{N}_0$ with $a_s > 0$, and suppose $H, J \in \text{Int}(A)$ satisfy*

$$HJ = \prod_{i=0}^s B_i^{a_i}.$$

There is an integer c with $0 \leq c \leq a_s$ such that

$$H/B_s^c \in \text{Int}(A), \quad J/B_s^{a_s-c} \in \text{Int}(A).$$

Both quotients split over K with all roots in V_{s-1} .

Proof. Unique factorization in $K[X]$ gives representations

$$H = C \prod_{f \in V_s} (X - f)^{u_f}, \quad J = D \prod_{f \in V_s} (X - f)^{w_f - u_f},$$

where $C, D \in K^\times$, $0 \leq u_f \leq w_f$, and w_f is the multiplicity of f in the numerator of $\prod_i B_i^{a_i}$. In particular,

$$w_f = a_s \quad (f \in V_s \setminus V_{s-1}). \quad (9)$$

Choose a monic irreducible polynomial $\pi \in A$ of degree s . Such a polynomial exists over every finite field in every positive degree. The set V_s is a complete system of representatives modulo π , and Lemma 2.1 gives

$$v_\pi(F_s) = 1, \quad v_\pi(F_i) = 0 \quad (0 \leq i < s).$$

Set $c = -v_\pi(C)$. Comparing leading coefficients in the identity for HJ gives

$$-v_\pi(D) = a_s - c.$$

For each $f \in V_s$, evaluate at $x = f + \pi$. Then $v_\pi(x - f) = 1$, while $v_\pi(x - g) = 0$ for every $g \in V_s \setminus \{f\}$. The integer-valuedness of H and J yields

$$u_f \geq c, \quad w_f - u_f \geq a_s - c. \quad (10)$$

For f in the nonempty set $V_s \setminus V_{s-1}$, equations (9) and (10) force $u_f = c$. Since $0 \leq u_f \leq a_s$ on this set, we have $0 \leq c \leq a_s$.

For all $f \in V_s$, the multiplicities $u_f - c$ and $w_f - u_f - (a_s - c)$ are nonnegative. They are zero on $V_s \setminus V_{s-1}$. We can consequently write

$$H = C\psi_s^c R,$$

where R has nonnegative root multiplicities and all roots in V_{s-1} . Regard R as a weighted product over V_s with zero weights on the outer set. Corollary 2.2 gives

$$H/B_s^c = CF_s^c R \in \text{Int}(A).$$

The same argument applied to J proves the second assertion. □

Proof of Theorem 1.1. The units of $\text{Int}(A)$ are exactly $\mathbb{F}_q^\times$. Indeed, a polynomial unit and its inverse must both have X -degree zero, and the constant elements of $\text{Int}(A)$ are exactly A .

Suppose $HJ = P$ with $H, J \in \text{Int}(A)$. We induct on the largest index s such that $a_s > 0$, omitting trailing zero exponents. If all exponents are zero, $HJ = 1$, and the assertion follows from the description of the units. If $s = 0$, unique factorization in $K[X]$ gives

$$H = CX^b, \quad J = C^{-1}X^{a_0-b}, \quad 0 \leq b \leq a_0.$$

Evaluation at $X = 1$ shows $C, C^{-1} \in A$, so $C \in \mathbb{F}_q^\times$.

Now let $s \geq 1$. Apply Lemma 3.1. The two resulting integer-valued quotients have product

$$\frac{H}{B_s^c} \frac{J}{B_s^{a_s-c}} = \prod_{i=0}^{s-1} B_i^{a_i}.$$

The induction hypothesis gives

$$H/B_s^c = u \prod_{i=0}^{s-1} B_i^{b_i}, \quad u \in \mathbb{F}_q^\times, \quad 0 \leq b_i \leq a_i.$$

Taking $b_s = c$ proves (3). Conversely, a polynomial of that form has an integer-valued complementary product, so it divides P .

For uniqueness, compare the multiplicity of any root in $V_s \setminus V_{s-1}$ in two expressions of the form (3). This determines b_s . Cancel this power of B_s and descend. At the final level, the multiplicity of zero determines b_0 . Equality of the remaining constants then determines u . $\square$

Proof of Corollary 1.2. The irreducibility of each B_i was proved in [2]; it also follows by applying Theorem 1.1 to $P = B_i$. If an irreducible polynomial divides a product P in that theorem, its expression (3) must contain exactly one basic factor, counted with multiplicity. Hence every irreducible divisor of P is associate to some B_i . Uniqueness of the exponent vector proves the assertion after taking $a_i = m\alpha_i$. $\square$

4 Consequences and examples

Corollary 4.1. *The submonoid*

$$\mathcal{C} = \left\{ u \prod_{i \geq 0} B_i^{a_i} : u \in \mathbb{F}_q^\times, a_i \in \mathbb{N}_0, \text{ finitely many } a_i \neq 0 \right\}$$

of $\text{Int}(A) \setminus \{0\}$ is divisor-closed. Its quotient by its units is the free commutative monoid on the classes of $B_0, B_1, \dots$

Proof. Divisor closure is exactly Theorem 1.1. The uniqueness statement in that theorem shows that the exponent-vector map identifies $\mathcal{C}/\mathbb{F}_q^\times$ with the direct sum $\bigoplus_{i \geq 0} \mathbb{N}_0$. $\square$

In particular, if $k = \sum_i \alpha_i q^i \geq 1$, then the divisor classes of β_k^m form a product of finite chains:

$$\{[H] : H \mid \beta_k^m\} \cong \prod_{i: \alpha_i > 0} \{0, 1, \dots, m\alpha_i\}.$$

This is also an isomorphism of partially ordered sets, with divisibility on the left and coordinatewise order on the right. Thus

$$\#\{[H] : H \mid \beta_k^m\} = \prod_i (m\alpha_i + 1), \quad \mathsf{L}(\beta_k^m) = \left\{ m \sum_i \alpha_i \right\},$$

where L denotes the set of factorization lengths. The uniqueness of factorization is stronger than the singleton assertion for lengths.

Example 4.2. For every $s > r \geq 0$, the index $k = q^s + q^r$ gives

$$\beta_k = B_s B_r.$$

Its m th power has exactly $(m+1)^2$ divisor classes, represented by $B_s^u B_r^v$ with $0 \leq u, v \leq m$. Its unique factorization has length $2m$. These mixed-index families have unbounded degree and are not covered by the prime-power assertion alone.

Remark 4.3. The analogue of Corollary 4.1 for the ordinary binomial polynomials fails. Put $b_n(X) = \binom{X}{n}$. Then

$$b_2(X)b_3(X) = X G(X), \quad G(X) = \frac{X(X-1)^2(X-2)}{12} \in \text{Int}(\mathbb{Z}).$$

The numerator of $G(x)$ is divisible by 3, as it contains three consecutive integers. It is also divisible by 4: for odd x the square $(x-1)^2$ suffices, while for even x the product $x(x-2)$ suffices. However, G is not a product of ordinary binomial polynomials, up to a unit. Any such product with roots contained in $\{0, 1, 2\}$ has weakly decreasing root multiplicities in the order 0, 1, 2, whereas those of G are (1, 2, 1). Thus products of the ordinary binomial polynomials need not form a divisor-closed monoid, even though each b_n is absolutely irreducible [1].

References

- [1] R. Rissner and D. Windisch, *Absolute irreducibility of the binomial polynomials*, Journal of Algebra **578** (2021), 92–114. doi:10.1016/j.jalgebra.2021.03.007.
- [2] R. Tichy and D. Windisch, *Irreducibility properties of Carlitz’ binomial coefficients for algebraic function fields*, Finite Fields and Their Applications **96** (2024), article 102413. doi:10.1016/j.ffa.2024.102413. Preprint: arXiv:2310.02061.
- [3] C. G. Wagner, *On the factorization of some polynomial analogues of binomial coefficients*, Archiv der Mathematik **24** (1973), 50–52. Author’s copy.